

PRODUCT REQUIREMENTS DOCUMENT

Comprint Unified Commerce Platform

Prepared for	Comprint Tech Solutions Pvt. Ltd.
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1. Executive Summary

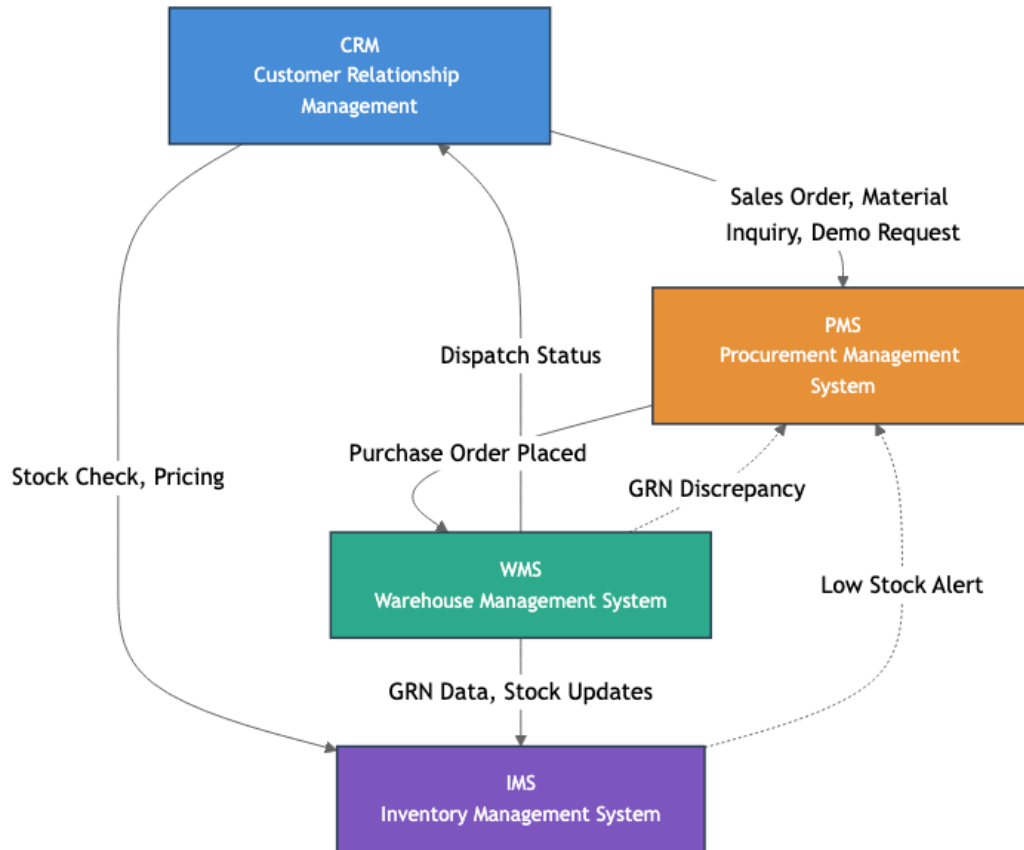
Comprint Tech Solutions manages IT hardware (servers, laptops, networking, storage, components) from customer interest to dispatch. Today this lifecycle is split across InvenTree (inventory), an old WMS (warehouse), and spreadsheets/WhatsApp for everything else — resulting in double-counted stock, untracked demos, zero traceability, and manual reconciliation.

This document defines requirements for a Unified Commerce Platform built on four interconnected systems:

System	Full Name	Owns
CRM	Customer Relationship Management	Sales pipeline, accounts, Material Inquiries, Demo Requests
IMS	Inventory Management System	Master data, categories, parts, stock, companies, pricing
PMS	Procurement Management System	Purchase Requests, quotations, PM approval, POs, GRN
WMS	Warehouse Management System	Inward (6 types), processing pipeline, outward (5 types), returns

Core Flow: Lead/Deal → Material Inquiry → Quote → Closed Won → Sales Order → Purchase Request (if needed) → PM Approval → Purchase Order → GRN → Inward → Inspection → Repair/Paint/Spares → QC → Stock → Allocation → Outward QC → Dispatch

2. System Architecture



CRM: Leads, Deals, Accounts, Contacts, Material Inquiries, Quotes, Sales Orders, Demo Requests

IMS: Categories (2-level), Parts, Stock Variants, Companies, Pricing, BOM Tracking

PMS: Purchase Requests, Quotations, PM Approval, Purchase Orders, GRN Matching, Vendors

WMS: Inward Batches, Inspection, Repair, Paint, Spares, QC, Rack Placement, Outward, Dispatch, Returns

3. Stakeholder Roles

Cross-Application Roles

Role	Description
Admin	Global configuration, cross-module visibility, master system oversight. Same view across all 4 systems.
Superadmin	Full read/write, user management, system settings. Present in every application.

System-Specific Roles

Role	System	Responsibilities
Sales Representative	CRM	Manage leads, raise Material Inquiry, create quotes and SOs
Account Manager	CRM	Customer accounts, SO/dispatch tracking, demo management
Product Manager (PM)	IMS + CRM	Category pricing, approve PRs and Demos per assigned categories
Procurement Manager	PMS	Oversee procurement, compare quotations, manage vendors, budget caps
Procurement Executive	PMS	Create PRs, collect quotations, create POs, track deliveries
Store Manager	IMS + WMS	Master data, maker-checker verification, stock accuracy, assembly/disassembly oversight
MIS (Inward Specialist)	WMS	Receive shipments, create batches, scan barcodes, GRN validation
Inspection Engineer	WMS	Run category-specific checklists, flag repair/paint/spares
Repair Technician	WMS	L2/L3 repairs, component replacement, BOM updates
QC Engineer	WMS	Inward QC (grading A/B), outward QC (dispatch clearance)
Dispatch Coordinator	WMS	Allocate stock, packing, vehicle assignment, challan generation
Operations Head	WMS + IMS	Cross-WMS visibility, escalation handling, operational approvals

Access Control (RBAC)

Permissions are not just role-based — they are filtered across four dimensions:

Dimension	Meaning	Example
Category	User sees/manages only assigned L1/L2 categories	PM for Laptops cannot approve Server PRs
OEM	User sees/manages only assigned manufacturers	PM assigned Dell + HP sees only those OEM parts and pricing
Variant	User restricted to New, Refurbished, or both	Refurbished warehouse team cannot access New stock
Tag	User can filter/act on specific tags	Spares engineer sees only NPL-tagged items

A user's effective permissions = Role × Category × OEM × Variant × Tag. Admins and Superadmins bypass all filters.

PM Assignment Granularity: Product Managers are assigned per Category × OEM combination (e.g., one PM for Dell Servers, another for HP Servers). A PM can also cover an entire category across all OEMs. This drives pricing authority, PR approval routing, and demo approvals.

4. CRM — Sales Pipeline

CRM manages sales from first contact to confirmed order. Creates structured requests (SO, Material Inquiry, Demo, PR) and sends them to backend systems. CRM stays lightweight — backend handles BOM, routing, complexity.

4.1 Lead Management

- **Stages:** New → Procurement (Material Inquiry raised) → Cold → Proposal → Negotiation → Closed Won / Closed Lost
- Kanban board + sortable list view
- **Fields:** TBD (customizable per deployment)
- **Closed Won trigger:** When a Quote is accepted and SO is created, the Lead/Deal moves to Closed Won and an Account is auto-created.

4.2 Deal Management

- **Stages:** Discovery → Proposal → Negotiation → Closed Won / Closed Lost
- Linked to Account, Lead (origin), Categories
- **Fields:** TBD (customizable per deployment)

4.3 Material Inquiry (Pre-Sales Feasibility)

A non-committal check raised by Sales before quoting the customer. Threaded conversation between Sales, Procurement, and PM.

- **Status:** Draft → Submitted → Partially Responded → Fully Responded → Closed
- **Links:** Lead (pre-deal) or Deal (mid-negotiation)
- **Fields:** TBD (customizable). Core fields: items, category, qty, budget, timeline.
- **Responses:** per item — qty available, price, available date, notes
- **Messages:** threaded with role tags (sales / procurement / PM)
- Split responses supported (e.g., 30 from stock + 20 fresh order)

4.4 Quotes

- **Status:** Draft → Sent → Accepted / Rejected / Expired
- **Fields:** TBD (customizable). Core: line items with part/description, category, qty, rate, amount.
- **Multiple quotes per inquiry:** A single Material Inquiry or Deal can have multiple competing quotes (different configs, OEMs, or pricing tiers).
- **Clone option:** Any existing quote can be cloned — copies all line items for quick revision.
- Versioning via parentQuoteld (revision chain)
- GST calculation, PDF preview
- Created from Material Inquiry response data when available

4.5 Sales Orders

- **Status:** Draft → Confirmed → Approved → Shipped → Delivered → Cancelled (approval merged into main status flow)
- **Approval:** [TO BE CONFIRMED: approver role — PM? Operations Head? Value-based?]
- **Fields:** TBD (customizable). Core: line items with category, variant, quantity, rate.
- **Fixed pricing:** Once an SO is generated, all line item pricing is locked. Backend cost changes do not affect existing SOs.

- **BOM visibility:** SO detail view displays BOM breakdown — which components are requested inside each assembly. Gives approver full visibility.
- **Component line items:** For customized orders, specific components (RAM, SSD, etc.) can be selected per part. Generates a Work Order for warehouse assembly.

Four fulfillment paths:

Path	When	Flow
Inventory-led	Stock available, standard config	SO → Allocate → Pick/Pack → Dispatch
Customized	Stock available, non-standard config	SO → Allocate → Work Order → Assembly/Disassembly → Outward QC → Dispatch
Procurement-led	Stock insufficient	SO → PR → Quotations → PO → Inward → ... → Allocate
Split order	Partial stock	Available items allocate immediately. Remaining → PR. SO stays "Partial".

4.6 Accounts and Contacts

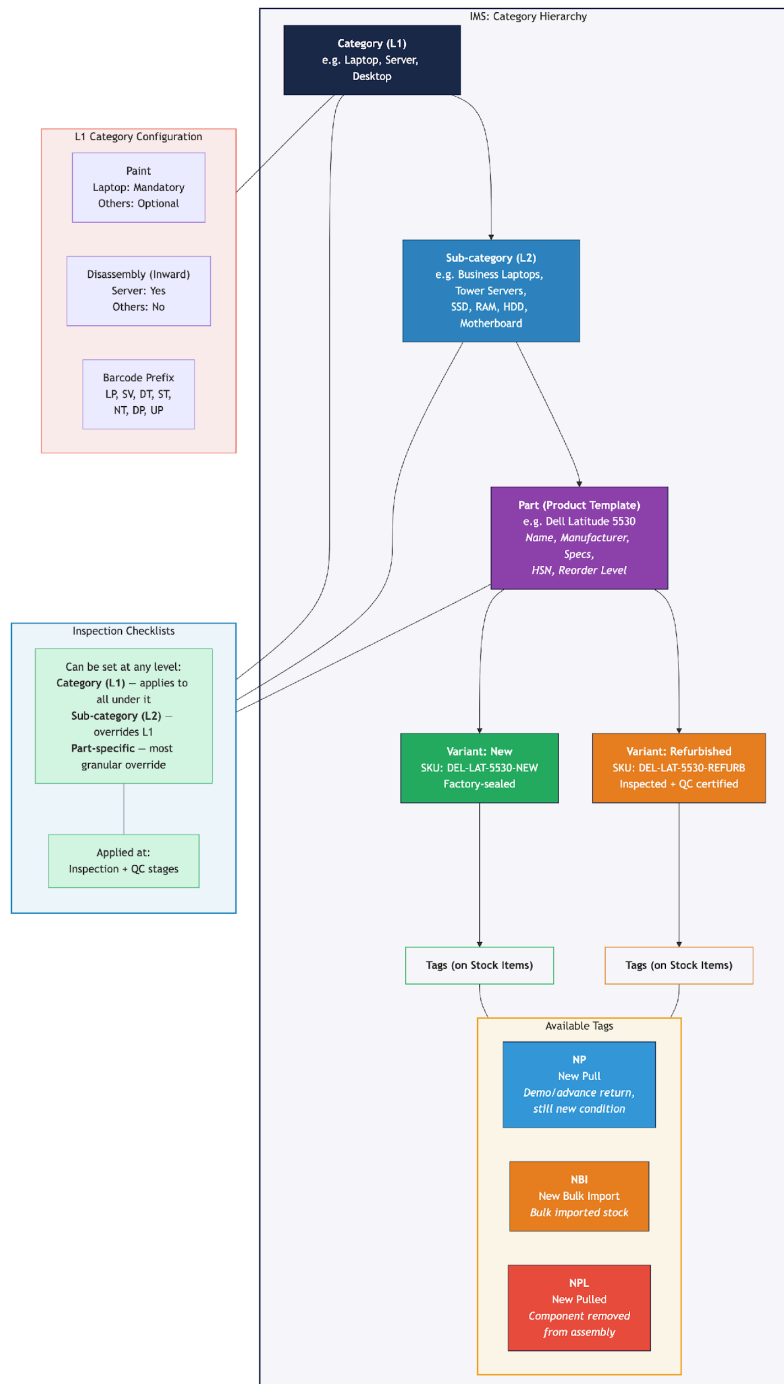
- **Account:** Auto-created when a Deal reaches Closed Won. Fields TBD (customizable).
- **Contact:** Person linked to Account. Fields TBD (customizable).

5. IMS — Inventory and Pricing

IMS is the master data backbone. Defines what parts exist, how they are organized, stock levels, and pricing.

5.1 Category Hierarchy

Two-level tree: Category (L1) → Sub-category (L2). Example: Laptop → Business Laptops. Sub-categories also include component types: SSD, RAM, HDD, Motherboard.



Category	Paint	Disassembly (Inward)	Prefix
Laptop	Mandatory	No	LP
Server	Optional	Yes	SV
Desktop	Optional	No	DT
Storage	Optional	No	ST
Networking	Optional	No	NT
Display	Optional	No	DP
UPS/Power	Optional	No	UP

Product Managers are assigned per Category × OEM combination (e.g., one PM for Dell Servers, another for HP Servers). A PM can also cover an entire category across all OEMs. This drives pricing authority, PR approval routing, and demo approvals.

Inspection Checklists: Can be configured at Category (L1), Sub-category (L2), or Part level. More specific levels override parent. Applied at Inspection and QC stages.

5.2 Parts (Product Templates)

Field	Description
Identity	Name, aliases, auto-generated SKU per variant, category, subcategory
Manufacturer	Brand, model
Specifications	Key-value pairs (RAM: 16GB, Storage: 512GB SSD)
Procurement	Reorder level, unit of measure, HSN code
Custom Fields	Admin-configurable dynamic fields (e.g., Tax %, Memory, CPU). Key-value pairs extensible without schema changes.
Checklists	Linked inspection/QC templates per category
Relationships	Compatible parts (BOM-native), Alternative parts (substitutes), Upgradable parts (swap options)

Auto-SKU: Part "Dell Latitude 5530" in Laptops → DEL-LAT-5530-NEW, DEL-LAT-5530-REFURB

5.3 Stock Management

Two structural variants (drive SKU generation, warehouse segregation, and PO controls):

Variant	Description
New	Factory-sealed items
Refurbished	Inspected, repaired, quality-certified

Tags (applied to stock items — extensible, admin-configurable, per D2):

Tag	Meaning	When Applied
NP (New Pull)	Returned from demo/advance return, still new condition	Auto-tagged on DEMO_RETURN and ADVANCE_RETURN inward
NPL (New Pulled)	Component removed from an assembly	Auto-tagged on BOM disassembly
NBI (New Bulk Import)	Bulk-imported stock	Applied during bulk import
Box Pack	Factory-sealed packaging intact	Manual or import
Local	Locally sourced item	Manual
(Custom)	Admin-defined tags	Manual

A demo return is variant = New + tag = NP — the manufacturing origin does not change because it went out for demo. Tags layer additional context without multiplying variants or SKUs.

Live Stock Counting: COUNT(stock_items WHERE part=[Part] AND variant=[Variant] AND status=IN_STOCK AND parent_item_id IS NULL). Components inside assemblies excluded. Never stored as a number.

5.4 Companies and OEMs

Companies — Customers and Suppliers:

One record per company with role flags: is_customer, is_supplier. Sub-resources: Contacts and Addresses. A company can be both customer and supplier simultaneously.

OEMs (Manufacturers) — Tracked separately from companies:

OEMs are the brands that manufacture products (Dell, HP, Cisco, Lenovo). Managed as an independent master list, not a company flag. Tracked model-wise — the system knows which OEM makes which parts.

Field	Description
Name	OEM brand name (Dell, HP, Cisco)
Models	Parts linked to this OEM as manufacturer
Categories	Which L1/L2 categories this OEM operates in
PM Assignment	Which Product Manager(s) manage this OEM

Why separate? OEMs don't have contacts, addresses, POs, or billing. They are a product classification axis — used for RBAC filtering, pricing rules, and procurement capping.

5.5 Pricing

- **Access:** Managed by Product Managers per assigned categories and OEMs (RBAC-filtered)
- Set/update pricing for parts within their categories
- Price history and cost trend visibility
- Pricing feeds into CRM quotes and SOs
- **Tags ↔ Pricing:** [TO BE CONFIRMED] Tag-based pricing differentiation — do tags like NPL, Box Pack affect price?

5.6 BOM Template and Component Relationships

When a PM or Admin creates a base part (server, laptop, storage device), they define its BOM relationships — the components that can go inside it. This replaces creating infinite variant combinations.

Relationship	Meaning	Example
Compatible	Components that fit natively	Dell Latitude 5530 → DDR4 16GB SODIMM, 512GB M.2 NVMe
Alternative	Substitutes if primary unavailable	Samsung 870 EVO → WD Blue SN570 (same form factor)
Upgradable	Swap options for customization	Server RAM: 16GB → 32GB → 64GB → 128GB

Cross-category mapping: A single component part (e.g., 512GB SSD) can be mapped as compatible/upgradable to multiple parent parts across categories.

Independent sale: Components can exist and be sold as standalone inventory without needing a parent part.

Archetype	Example	BOM Behavior
Fixed part with variations	Laptop (MacBook M1)	Components mapped. Assembly rare but supported if upgradable.
Composite part	Server (ProLiant DL380)	Multiple components inside. SO-driven assembly/disassembly is common.

Standalone component	Storage (512GB SSD)	IS itself a component. Can exist independently OR inside a parent part.
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5.7 BOM Operations and Assembly

Oversight: The Store Manager oversees all assembly and disassembly operations.

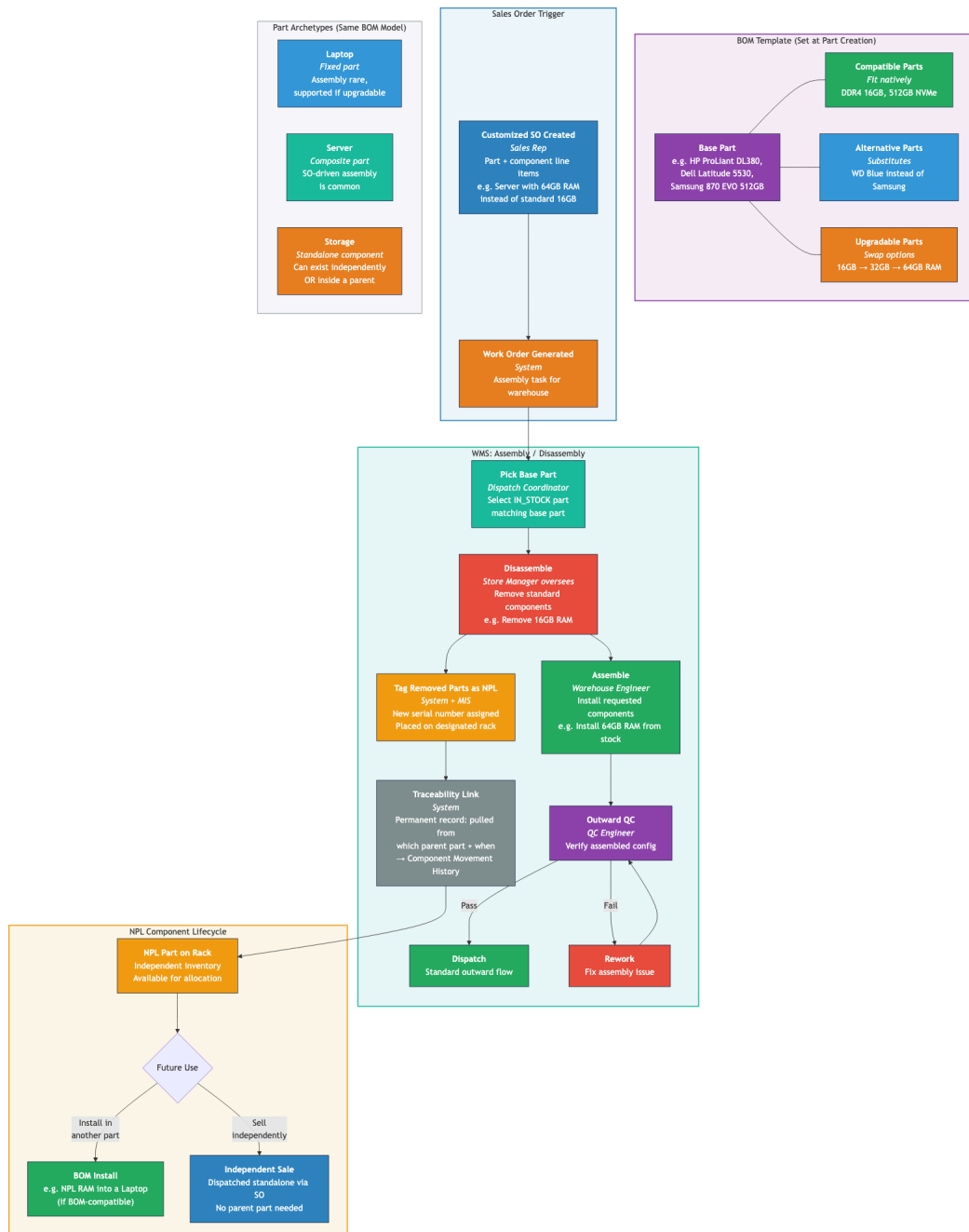
Operation	What Happens	Stock Impact	Performed By
Install	Component's parent_item_id → parent part	Disappears from free stock	Warehouse Eng., verified by Store Manager
Remove	parent_item_id cleared. Tagged NPL. New serial.	Reappears as free stock	Warehouse Eng., verified by Store Manager
Swap	Remove old (NPL) + install new (consumed)	Old → free, new → consumed	Warehouse Eng., verified by Store Manager

Re-inwarding pulled components: When removed, component is tagged NPL, assigned new serial, placed on rack, recorded in Component Movement History table.

5.8 Component Movement History (Traceability)

Every component maintains an unbroken chain of custody. Stored in a dedicated relational table, not just audit logs.

Field	Description
movement_id	Unique ID
item_id	The component (stock item) being moved
movement_type	INSTALLED / REMOVED / SWAPPED / TRANSFERRED
parent_item_id	Assembly this component was installed in or removed from
work_order_id	Work Order that triggered this movement
from_rack_id / to_rack_id	Rack the component moved from/to
performed_by / verified_by	User who executed + Store Manager who verified
timestamp + notes	When and free-text context



Tags are dynamic and extensible: NPL (New Pulled), NP (New Pull), NBI (New Bulk Import), Box Pack, Local, and admin-configurable custom tags. Applied to stock items, searchable and filterable.

6. PMS — Procurement

PMS manages the full procurement lifecycle from purchase need to goods receipt.

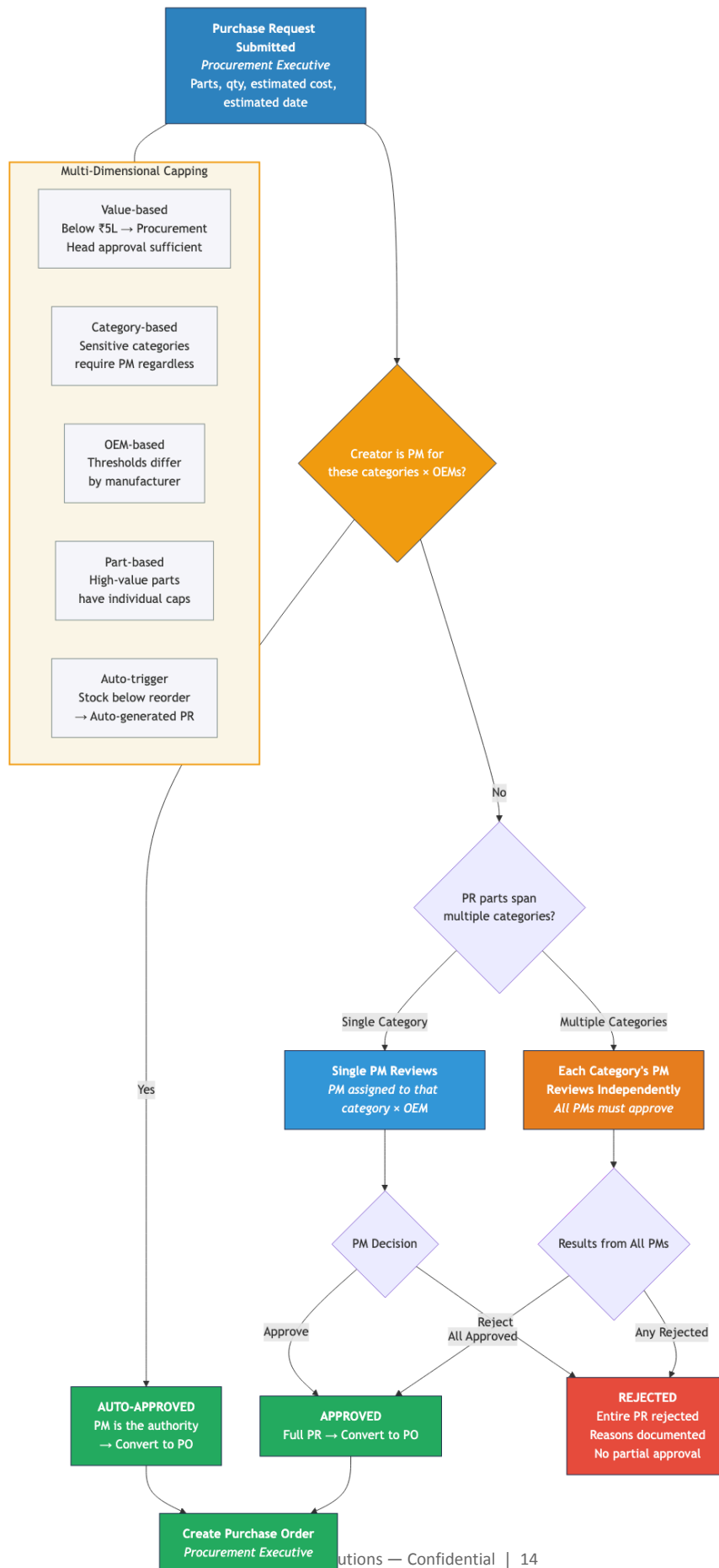
6.1 Purchase Requests

Field	Description
Status	Draft → Submitted → Under Review → Approved → Rejected → Converted to PO
Items	Part (optional — can be description-based), category, qty, estimated cost
Estimated Date	When items are needed — replaces urgency levels. Drives procurement prioritization.
Links	Material Inquiry (if feasibility checked), SO (if customer-linked), auto-trigger (stock below min)
Justification	Free-text: why this purchase is needed

Description-based line items: PR items can reference a master Part OR be a free-text description for ad-hoc procurement.

PM Auto-Approval: If the PR creator is the PM assigned to the relevant categories/OEMs, the PR is auto-approved — no separate approval step needed.

Approval Logic:



- PM approval per category × OEM. All PMs must approve for PR to proceed.
- All approved → APPROVED → Convert to PO
- Any rejected → entire PR REJECTED with reasons (no partial approval)

Multi-dimensional capping:

Dimension	Rule
Value-based	Orders below ₹5L → Procurement Head approval sufficient
Category-based	Sensitive categories → PM approval regardless of value
OEM-based	Thresholds can differ by manufacturer
Part-based	Specific high-value parts can have individual caps
Auto-trigger	System generates PR when stock below reorder level

6.2 Quotation Collection

- **Minimum 3 quotations** per approved PR (Local or International suppliers)
- Side-by-side comparison: price, lead time, payment terms

6.3 Purchase Orders

Field	Description
Status	Draft → Sent to Vendor → Acknowledged → Partially Received → Fully Received → Reconciled → Closed → Cancelled
Line Items	Part (or description-based), qty ordered, qty received, unit price, tax rate, amount, delivery date
Financials	Subtotal, tax, discount, grand total
Terms	Payment (Net 30/60), Delivery (FOB/CIF/Ex-Works)
Variant Control	New or Refurbished — controls which warehouse can receive this PO

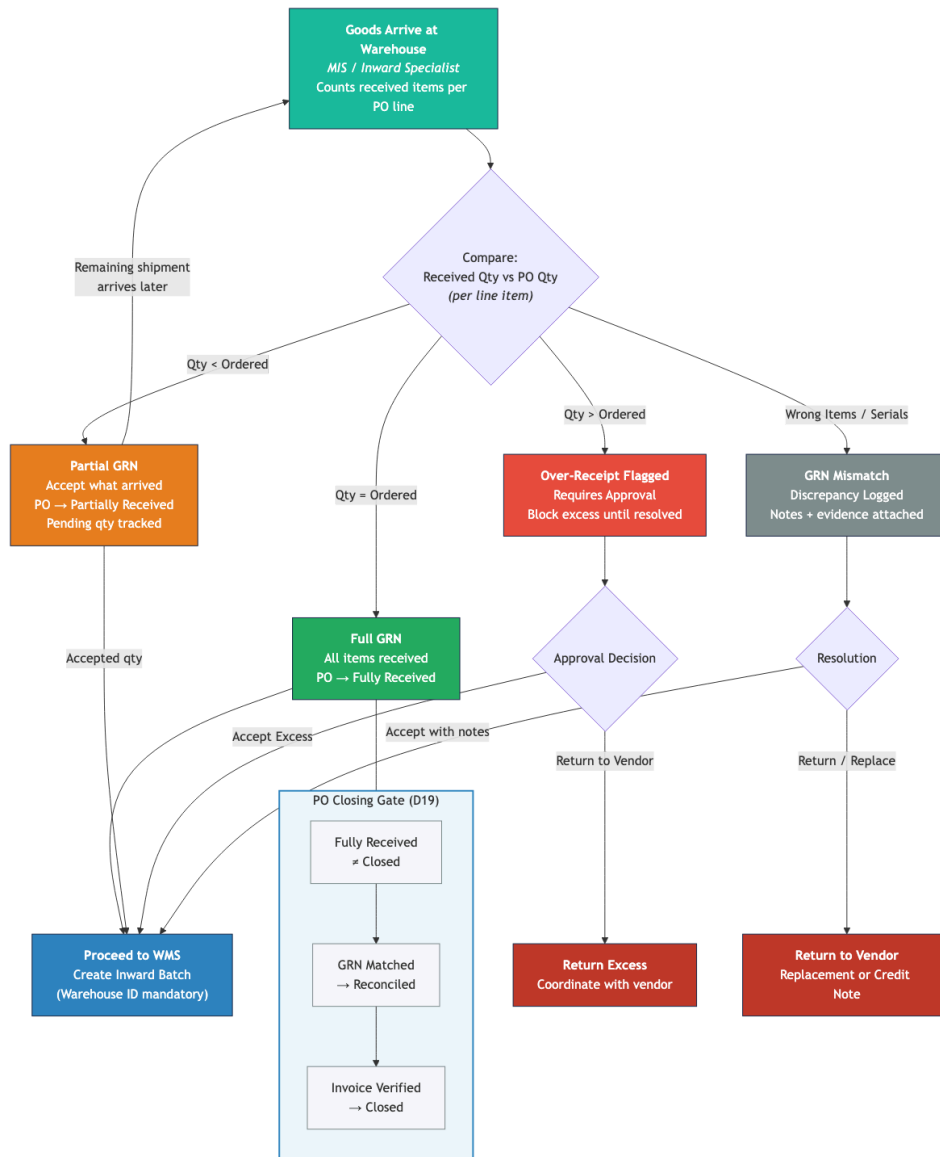
Mark as Received: Each line item tracks qty_ordered vs qty_received. Deliberate user action per line item.

Status Logic: Received=0 → Open | Received<Ordered → Partially Received | Received=Ordered → Fully Received | GRN matched → Reconciled | Invoice verified → Closed

Closing gate: PO only reaches Closed when all items received (GRN matched) AND invoice amounts reconcile.

6.4 GRN Matching

GRN validates received quantities against PO line items. Per-line-item matching.



Scenario	Action
Qty = PO	Accept Full GRN. PO → Fully Received. Proceed to WMS.
Qty < PO	Partial GRN. Accepted qty proceeds to WMS. Pending qty tracked. PO → Partially Received.
Qty > PO	Block OR Approval Workflow. Over-receipt flagged.
Wrong items / serials	GRN Mismatch logged. Resolution: Return/Replace to Vendor OR accept with notes.

6.5 Vendor Management

- **Fields:** name, code, contact person, GST, PAN, bank details
- **Performance:** rating (1-5), on-time delivery %, quality score %
- **Status:** Active / Inactive / Pending Approval / Suspended / Blacklisted

7. WMS — Warehouse Operations

WMS manages the physical lifecycle of every asset from arrival to exit.

7.1 Inward (6 Types)

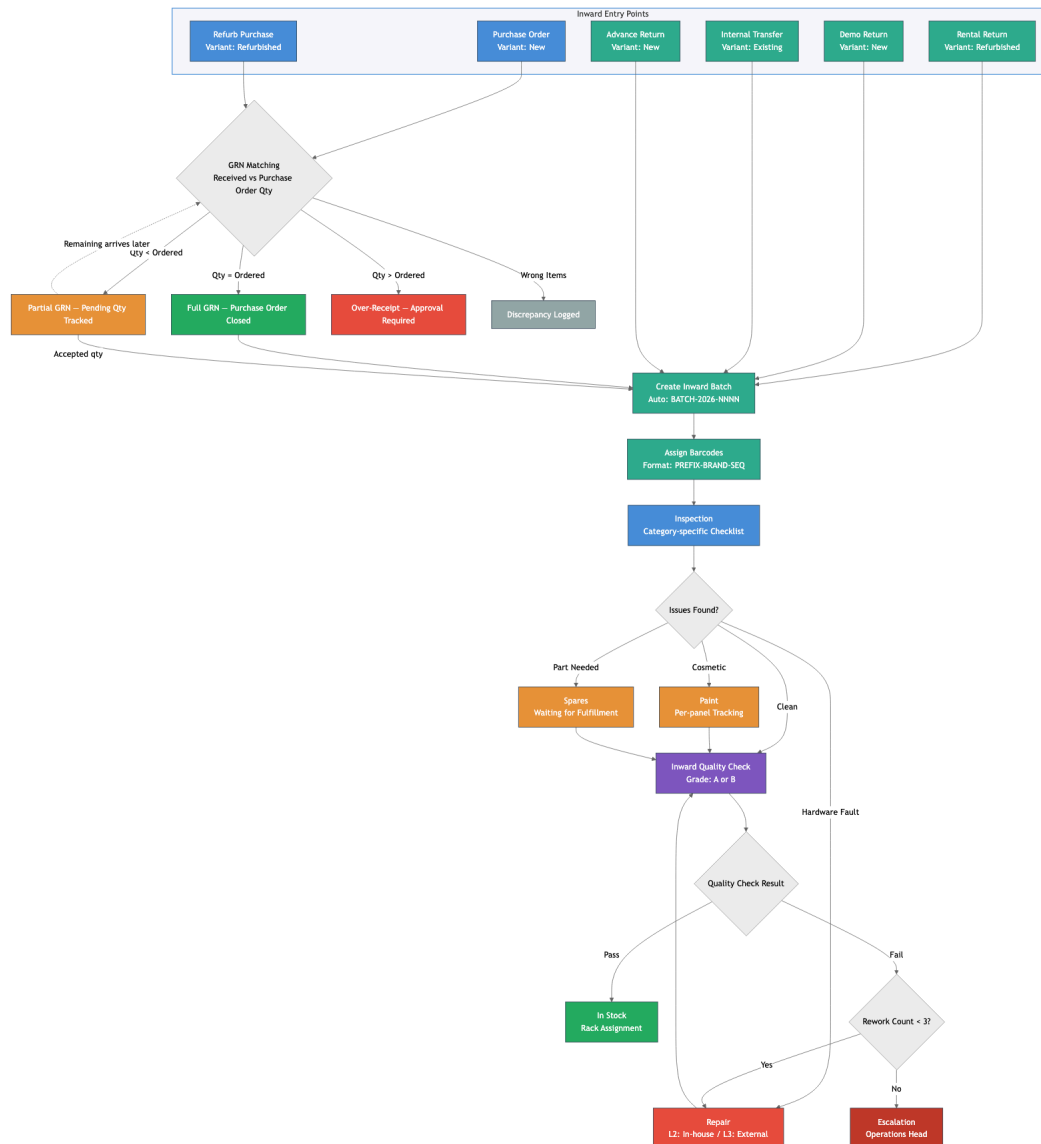
Type	Source	Stock Variant	Auto-Tag	PO Required
PURCHASE_ORDER	Vendor (new goods)	New	—	Yes
REFURB_PURCHASE	Vendor (refurbished)	Refurbished	—	Yes
RENTAL_RETURN	Customer (rental ended)	Refurbished	—	No
DEMO_RETURN	Customer (demo completed)	New	NP	No
INTERNAL_TRANSFER	Inter-warehouse / department / HR MS	(keeps existing)	—	No
ADVANCE_RETURN	Customer (contract ends)	New	NP	No

- **Auto number:** BATCH-2026-NNNN
- **Mandatory Warehouse ID:** Every inward batch must capture which warehouse the items are received into. No inward without warehouse assignment.
- **Inward source linking:** PO link mandatory for PURCHASE_ORDER and REFURB_PURCHASE. All other type-specific linking (Demo Return → customer, Internal Transfer → employee/HR MS, etc.) is [TBD].
- **PO-to-Warehouse Control:** System only shows POs matching warehouse variant type. Refurbished PO invisible to New Stock warehouse.
- **Barcode format:** {prefix}-{brand}-{sequence} (e.g., LP-DEL-4521)
- When PO selected, auto-fetch: items, ordered qty, vendor details, expected delivery

HR MS Integration: INTERNAL_TRANSFER supports integration with HR Management System portal. Employee hardware assign/return flows through the same WMS pipeline.

7.2 Processing Pipeline

All 6 inward types enter the same quality pipeline.



Inspection: Category-specific digital checklist. Groups: Panels, Display, Input, Audio, Power, Hardware, Ports. Per unit: Pass/Fail/N/A. Failures trigger routing flags.

Inspection Engineer Assignment: Auto-assignment: round-robin within engineers for that category, balanced by workload. Manual override by Store Manager / Ops Head. Stock items in a batch can be split across engineers.

Repair: L2 (in-house) or L3 (external). TAT tracked: green (<3d), yellow (3-5d), red (>5d). L2 → L3 escalation. Component replacement updates part BOM.

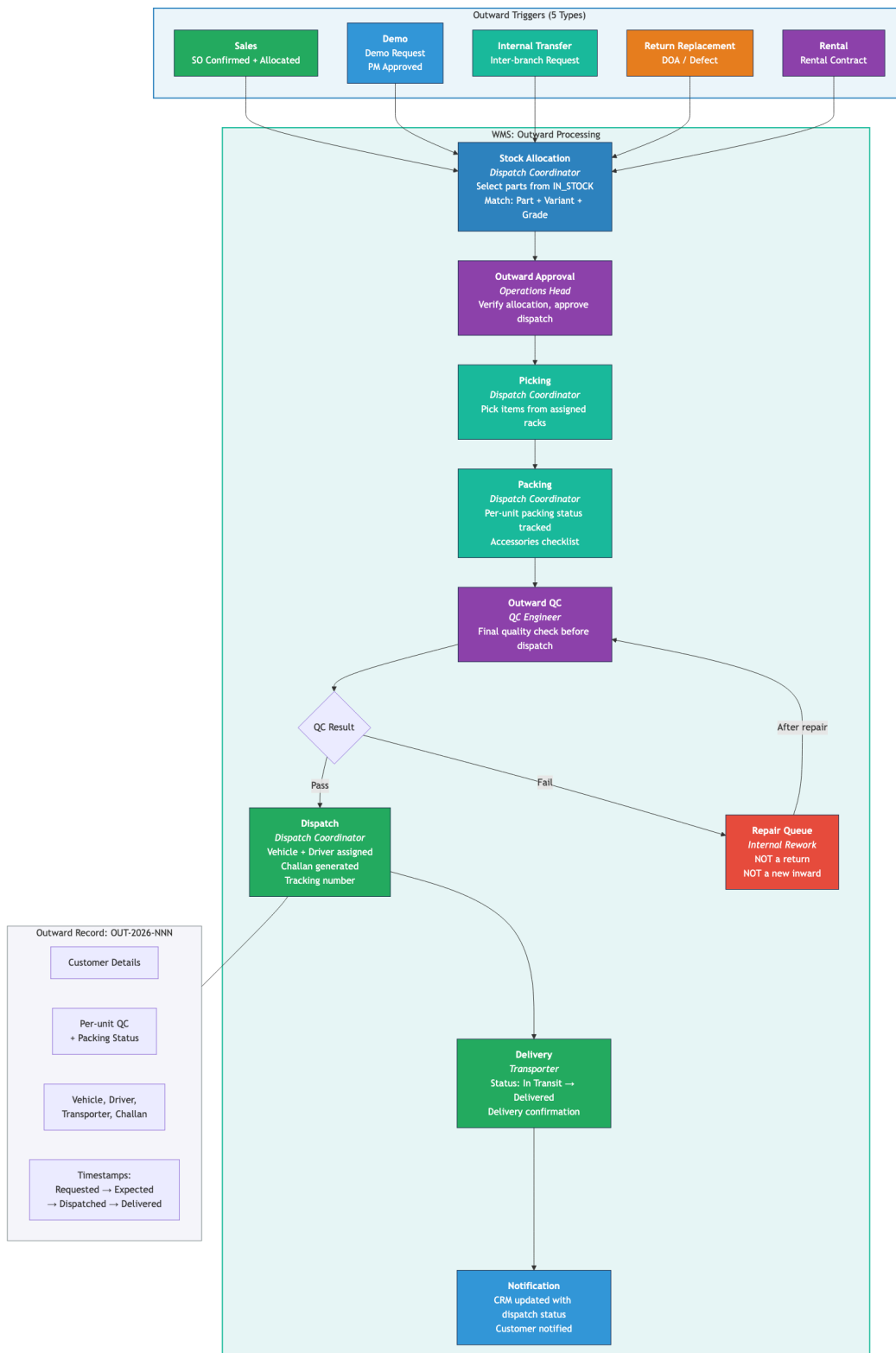
Paint: Mandatory for Laptops (L1 config). Per-panel: Top Cover, Bottom Cover.

Spares: Stock item blocked (WAITING_FOR_SPARES) until spare fulfilled from stock. Spares tracked with rack-level precision — designated spares racks, location-wise visibility.

QC: Configurable grading system — default A/B, admin can add grades over time (e.g., A+, A, B, C, Scrap). Failure → repair queue. Max 3 rework attempts, then escalation.

7.3 Outward (5 Types)

Type	Trigger	SO Link
SALES	SO confirmed + allocated	Yes
DEMO	Demo Request approved by PM	No (demo link)
INTERNAL TRANSFER	Inter-branch request	No
RETURN REPLACEMENT	DOA / defect replacement	Yes (original SO)
RENTAL	Rental contract	No (contract link)



Standard dispatch: Approval → Picking → Packing → Outward QC → Dispatch → Delivery

Customized SO: Allocate base part → Work Order → Disassemble (remove standard components, tag NPL) → Assemble (install requested parts) → Re-inward pulled parts → Outward QC → Dispatch

Outward QC Failure: Stock item routes to repair queue — internal rework, NOT a return. Re-enters outward QC after repair.

Outward Record: Auto number (OUT-2026-NNN), mandatory Warehouse ID, customer details, per-unit QC/packing status, logistics (vehicle, driver, transporter, tracking, challan), timestamps.

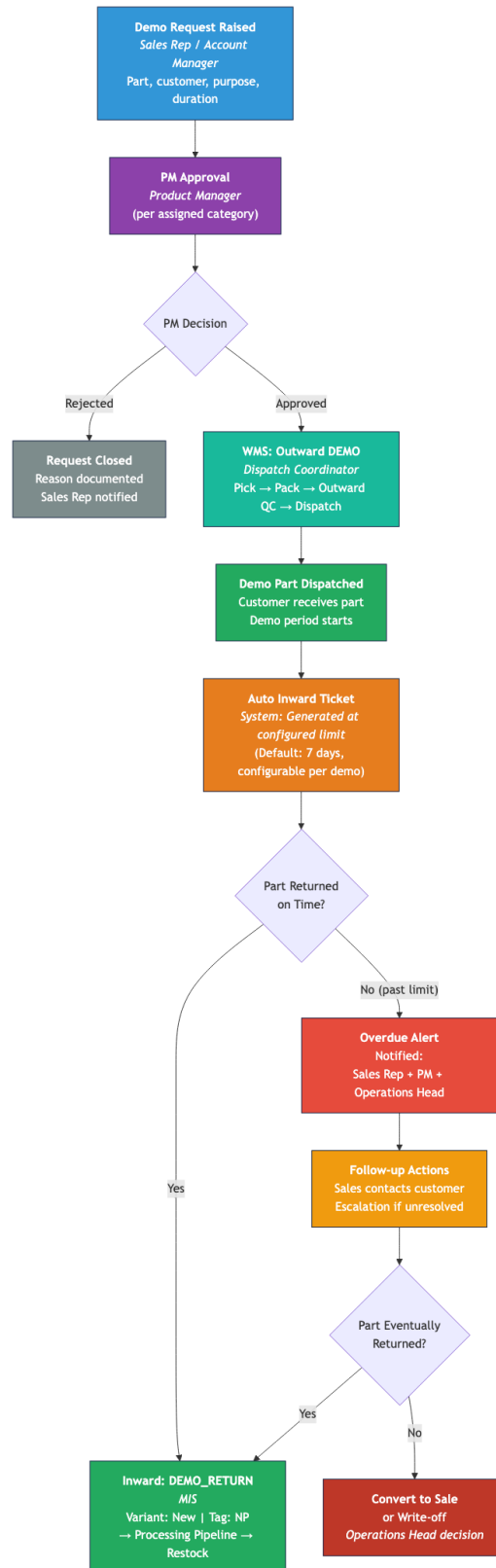
7.4 Returns

Reason	Action
QC_FAILED	Internal: route to repair → re-enter outward QC
CUSTOMER_RETURN	Return pickup → inspect → repair or restock
DAMAGE_IN_TRANSIT	Inspect → repair or scrap. File transporter claim.
WRONG_ITEM	Restock correct item, dispatch the right one

Note: Customer Replacement and Customer Return are the same flow (CUSTOMER_RETURN). Additional return types (Advance Replacement, Wrong ID / Item Mismatch) are [TBD].

Flow: Initiated → Received at warehouse → Inspected → Resolved (per unit: Repair / Restock / Dismantle)

7.5 Demo Lifecycle



8. Inventory States

State	Meaning
Received	GRN done, barcode assigned
Under Inspection	Checklist in progress
In Repair	Repair job active
Waiting for Spares	Blocked until spare fulfilled
In Paint	Paint job active
QC Pending	Awaiting quality check
Ready for Stock	QC passed, pending rack assignment
In Stock	Rack assigned, available for allocation
Allocated	Reserved for SO
Dispatched	Shipped out
Scrapped	Marked for end of life — routes to Dismantle
Dismantle	Component recovery from scrapped items (BOM disassembly). Usable parts re-enter stock as NPL.
Disposed	Final disposal after dismantling. No recoverable value.

9. Cross-Module Integration

From	To	Trigger	Data
CRM	PMS	Material Inquiry created	Parts, qty, budget, timeline
CRM	PMS	SO confirmed, stock insufficient	SO → triggers PR
CRM	WMS	Demo approved	Part spec, customer, return date
PMS	WMS	PO placed	PO, parts, variant type, supplier
PMS	IMS	PR approved	Part demand signal
WMS	IMS	Stock item created (maker-checker)	Part, variant, serial, barcode
WMS	IMS	QC passed	Stock item, grade, rack location
WMS	IMS	BOM change	Component swap, tag update
CRM	WMS	Customized SO	Work Order: part + component mods
WMS	IMS	NPL re-inward	Pulled component: new serial, rack, origin link
IMS	PMS	Stock below minimum	Part, qty → auto PR
WMS	CRM	Dispatched	SO, tracking, delivery ETA
System	Sales+PM+Ops	Demo overdue	Part, customer, dispatch date
System	PMS	GRN discrepancy	PO, part, qty mismatch

10. End-to-End Flow Summary

Complete procurement-led sale: CRM → PMS → WMS → IMS with role annotations at each step.

Step	System	Action	Role
1	CRM	Lead/Deal → Material Inquiry	Sales Rep → Procurement → PM
2	CRM	Quote Created (from Material Inquiry data)	Sales Rep
3	CRM	Closed Won → Account Created	Sales Rep
4	CRM	Sales Order Confirmed (pricing locked)	Sales Rep
5	PMS	Purchase Request (if needed)	Procurement Executive
6	PMS	PM Approval (per category × OEM)	Product Manager
7	PMS	Purchase Order → Vendor	Procurement Executive
8	PMS	GRN Matching → Proceed to WMS	MIS validates qty
9	WMS	Inward Batch + Barcodes (Warehouse ID mandatory)	MIS
10	WMS	Inspection → Repair / Paint	Inspection Eng. → Repair Tech
11	WMS	QC (Configurable grades)	QC Engineer
12	WMS	IN_STOCK → Allocate → Outward QC	Dispatch Coordinator → QC
13	WMS	Pack → Dispatch → Delivery	Dispatch Coordinator

11. Key Design Decisions

#	Decision	Rationale
D1	Material Inquiry before Quote/SO	Sales needs real pricing before committing. Material Inquiry is non-committal.
D2	Tags over nested variants	Only 2 structural variants (New, Refurbished). NPL/NP/NBI are tags — extensible without schema changes.
D3	PO carries variant type	Prevents cross-contamination: New PO invisible to Refurb warehouse.
D4	PM approval per category × OEM	Multi-PM if PR spans categories. Full approval or rejection — no partial. PM-created PRs auto-approved.
D5	CRM independent from backend	CRM sends structured requests. Backend handles complexity.
D6	Demo = approval + configurable limit	Approval gate, auto return ticket (default 7d, configurable), overdue escalation.
D7	Single pipeline for all inward	All 6 types go through same quality process.
D8	Stock = live COUNT	Never stored. BOM-aware: excludes installed components.
D9	Category config drives behavior	Paint, disassembly, allowed variants from L1 flags.
D10	Companies + OEMs separated	Companies = customers/suppliers. OEMs = manufacturers (RBAC/pricing axis).
D11	Outward QC fail → repair	Internal rework, not a return. Stock item re-enters outward QC.
D12	GRN matching per line item	Partial receipt, discrepancy docs, over-receipt flagging.
D13	Split SO fulfillment	Inventory-led + Procurement-led in same order.
D14	BOM as template	Parts define compatible/alternative/upgradable at creation time.
D15	SO-driven assembly	Customized SO → Work Order. Standard SO skips assembly.
D16	Cross-category components	One SSD can be compatible with both Server and Laptop.
D17	Components sellable independently	Pulled parts can be dispatched standalone — no parent part required.
D18	NPL re-inward with traceability	Pulled parts get new serial, rack, origin link.
D19	PO closes on reconciliation	Fully Received ≠ Closed. PO closes only after GRN + invoice reconciliation.
D20	Description-based PR/PO items	Line items can be master Parts OR free-text. Supports ad-hoc procurement.
D21	Estimated date over urgency	PRs use concrete "needed by" date instead of High/Medium/Low.
D22	Multi-dimensional capping	PR/PO thresholds by value + category + OEM + part.
D23	Component movement history	Dedicated relational table tracks every install/remove/swap with parent part, work order, rack, and performer.
D24	Configurable QC grades	Default A/B, admin can add grades. Dynamic grade list.

D25	Inspection engineer auto-assign	Round-robin by category + workload. Manual override available.
D26	HR MS integration	Employee hardware flows through same WMS pipeline via HR MS portal.
D27	Fixed SO pricing	Once SO is generated, pricing is locked. Backend cost changes don't affect existing SOs.
D28	Scrap → Dismantle	Scrapped items route to Dismantle for component recovery before final disposal.
D29	Mandatory Warehouse ID	Every inward/outward must capture which warehouse. No stock movement without warehouse.

12. Out of Scope (To Be Decided)

Area	Status
Billing, Invoicing, GST	TBD (Tally integration vs native)
Accounting (GL, AR, AP)	TBD
Reconciliation (Bank, Vendor)	TBD (GRN matching IS in scope. PO reconciliation with Tally IS in scope for closing gate.)
Rentals (contracts, billing)	TBD
Collections and Payments	TBD
Challan Generation	[TBC] Native WMS or pushed to Tally?
Service Desk (SDP)	Integration contract only
E-commerce	Future phase
AI Agents (DPO, Alisha)	Future phase
HR MS Integration	Integration contract defined. Internal transfers from HR MS portal.

13. Glossary

Term	Definition
BOM	Bill of Materials — components inside an assembled part
GRN	Goods Receipt Note — receipt record against a PO
MI	Material Inquiry — pre-sales feasibility check
MIS	Inward Specialist role title
NPL	New Pulled — tag for component removed from assembly (variant stays unchanged)
NP	New Pull — tag for returned item still in new condition (variant = New + tag = NP)
PM	Product Manager
PMS	Procurement Management System
PO	Purchase Order
PR	Purchase Request
QC	Quality Control
SKU	Stock Keeping Unit
SO	Sales Order
TAT	Turn Around Time

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